COLD PLASMA FOR DECONTAMINATION OF CHICKEN BREAST, ROMAINE LETTUCE, AND STAINLESS STEEL SURFACES FROM MULTI-DRUG RESISTANT *SALMONELLA ENTERICA* SEROVAR HEIDELBERG

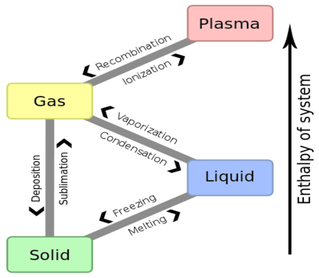
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Background

- In the United States, 1.2 million illnesses are caused by non-typhoidal salmonella resulting in 19,000 hospitalizations and 450 deaths, annually (CDC).
- Poultry meat is the major implicated food, but contaminated fresh produce is also implicated.
- Estimated economic cost somewhere between $2.3 and $11.3 billion annually.
- Of the 2500 serovars of Salmonella, *Salmonella enterica* serovar Heidelberg (**SH**) is most commonly found in chickens and turkeys.
- Antimicrobial drug resistance in SH is notable and is associated with increased risk of hospitalization of infected individuals.
- Increasing number of multi-drug resistant S. Heidelberg (**MDR-SH**) outbreaks linked to poultry have occurred recently in the US.

What is Cold Plasma?



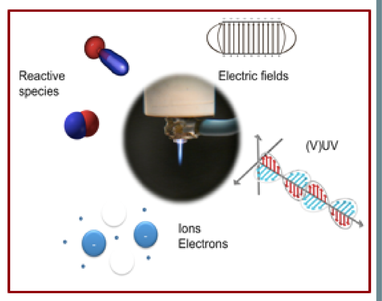
Plasma is the 4th state of matter. It is partially or completely ionized gas with a net neutral charge consisting of:

- Electrons
- Photons
- Neutral molecules
- Negative ions
- Positive ions &Radicals

Light species

Heavy species

The electric field accelerates electrons in the gas and their energetic collision with the surrounding gas leads to ionization, dissociation of molecules, and the formation of UV and reactive chemical species such as OH•, NO•, $^1O_2$, O_3, H_2O_2 and O_2-



Cold atmospheric gaseous plasma (**CAP**) is a non-thermal technology that may play a significant role in decontaminating food and food-contact surfaces from bacterial pathogens.

Aim of Investigation

- Studying the bactericidal efficacy of a 2D-air-based DBD-plasma microdischarge array (2D-AMPA) against an MDR-SH isolate on three different surfaces (stainless steel, romaine lettuce, and chicken breast).
- Understanding the factors affecting the bactericidal activity on each surface.

Methods

1- Cold Plasma generation setup:

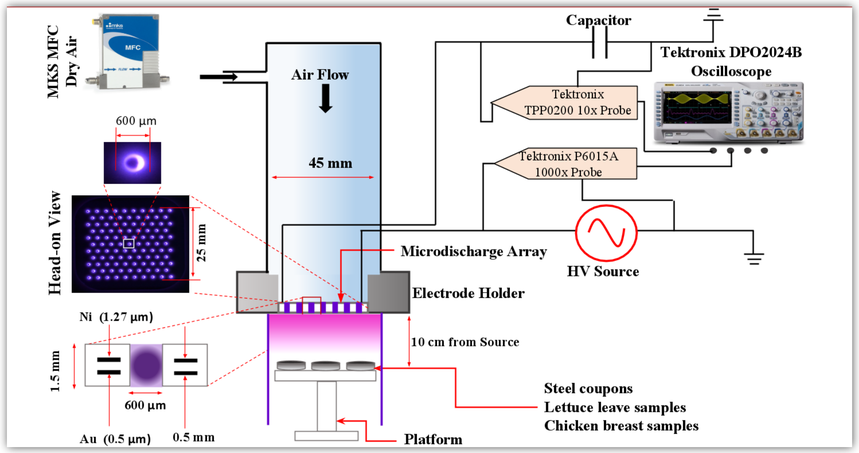


Figure 1: Schematic of the two dimensional air-based DBD-plasma microdischarge array (2D-APMA).

2- Bacterial Strain: Four drug resistant strains of *Salmonella enterica* serovar Heidelberg isolated from a chicken farm in Minnesota.

3- Samples: Stainless steel discs and squares (1 cm^2) of romaine lettuce leaves and chicken breast meat. Food samples were initially disinfected by sodium hypochlorite solution then rinsed with distilled water. Each sample was spiked with 20µL of stationery phase culture of SH grown in trypticase soy broth. The initial count was adjusted by McFarland standard. All samples were dried in a biosafety cabinet for approximately 20 minutes.

4- Operational conditions of cold plasma treatment:
Air flow rate: 16.4 SLM -- **Power:** 12.7 Watt -- **Exposure distance:** 10 cm
Wet exposure (water sprayed every 2 min.)

5- Enumeration of survived bacteria: Agar plate count technique using tryptic soy agar was used. The plates were incubated at 37ºC for 24hr. The count of survived bacteria was calculated by the difference between bacterial count in control and treated samples.

Results

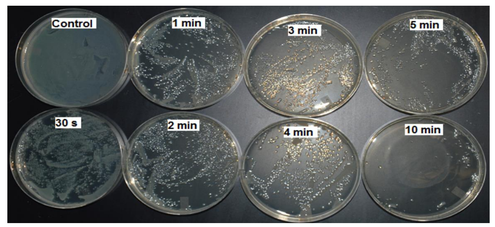


Figure 2: Bactericidal activity of CAP against MDR-SH on agar culture exposed at 12.5 cm to the 2D-AMPA treatment.

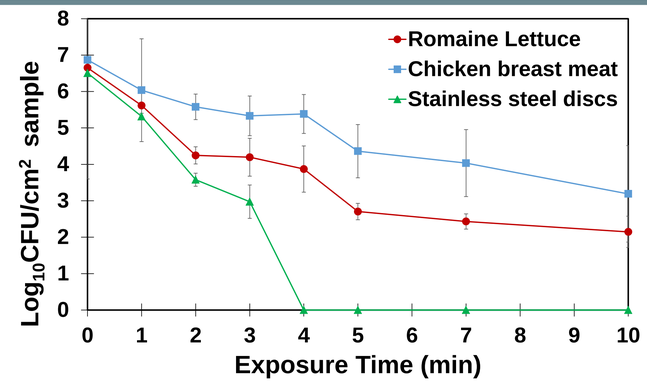


Figure 3: Comparison between the efficacy of CAP against MDR-SH on various food and food-contact surfaces.

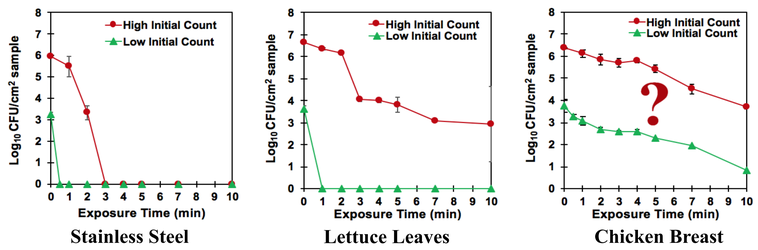


Figure 4: Bactericidal effect at low and high initial count of MDR-SH.

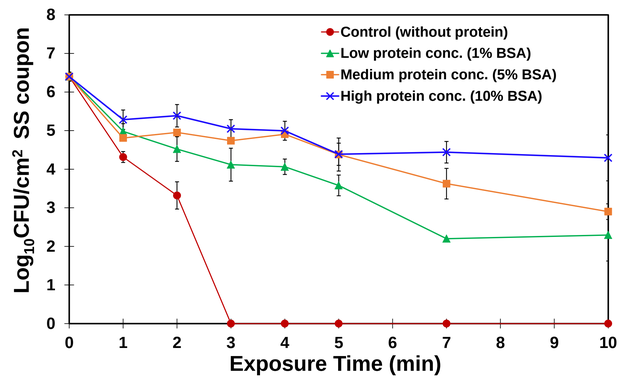


Figure 5: Suppression effect of proteins on CAP bactericidal activity.

Conclusion

- 2D-AMPA setup is cost-effective since it can be used for decontaminating surfaces using "air" at relatively low power.
- It showed a strong bactericidal activity against MDR-SH.
- Bactericidal activity of CAP depends on operational conditions e.g., initial bacterial load of treated surface, humidity, and topography and nature of the treated surface.

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